ROBOARM — FULL WIRING (REV E)

Every board pin shown. Wires drawn between parts. Power rails use KiCad power symbols (+24V, +5V, +6V, GND). Unused pins carry no-connect flags.

UART bus A: Mega Serial2 (TX2 D16 via 1k, RX2 D17) -> PDN/UART of DRV1–3 (addr 0..2).
UART bus B: Mega Serial3 (TX3 D14 via 1k, RX3 D15) -> DRV4–5. Gives per-joint load (StallGuard), temperature and open-wire flags.

Joint encoders: AS5600 reads a magnet on each joint OUTPUT shaft (after the gearbox), so the angle is absolute at power-up.
OUT = analog angle (0–5 V) to Mega A0–A4 through 1k + 100 nF at the Mega end. DIR tied to GND sets direction.
Check the module runs at 5 V (VCC) before buying; SDA/SCL left free for later I2C use. GPO tied to 5 V keeps OUT in analog mode.

E-STOP: pressing SW1 opens the relay coil circuit, K1 drops out
and the red 24 V bus goes dead. Pi and Mega stay on.
A broken E-stop wire also stops the motors (fail-safe).

PS1

Lab bench supply: 24 V, 3 A limit (later: Mean Well LRS-200-24)

F0

10 A main

K1

24 V 30 A relay

SW1

E-STOP (NC, latching)

PS1 for first tests: existing bench supply, 24 V, current limit 3 A.
Its output on/off button cuts motor power only; Pi and Mega stay on.
The 24 V goes ONLY to the motor drivers and the servo buck. The Pi has its own USB-C supply (PS2).

D0

1N4007

24 V BUS (red): bench supply + -> F0 -> VM on every driver. Ground return (black): every driver GND -> bench supply −

PS2

Official Pi 27 W USB-C supply

U1

Raspberry Pi 5 8GB

Pi USB-A to Mega USB-B cable:
powers the Mega and carries commands.

U2

Arduino Mega 2560 R3

R31

1k (UART-TX)

R21

10k

DRV1

TMC2209 module (StepStick)

M1

17HS19-1684S-PG5

J1 Base: 0.52 Nm 1.68 A × 5.1Ω:1, set 1.4 A RMS

C20

100uF 35V

R22

10k

DRV2

TMC2209 module (StepStick)

M2

17HS19-1684S-PG14

J2 Shoulder: 0.52 Nm 1.68 A × 13.73:1, set 1.4 A RMS

C21

100uF 35V

R23

10k

DRV3

TMC2209 module (StepStick)

M3

17HS19-1684S-PG5

J3 Elbow: 0.52 Nm 1.68 A × 5.1Ω:1, set 1.4 A RMS

C22

100uF 35V

R32

1k (UART-TX)

R24

10k

DRV4

TMC2209 module (StepStick)

M4

17HS08-1004S

J4 Wrist: 0.16 Nm 1 A, set 1.0 A RMS

C23

100uF 35V

R25

10k

DRV5

TMC2209 module (StepStick)

M5

11HS12-0674S

J5 Wrist twist: 0.06 Nm 0.67 A, set 0.67 A RMS

C24

100uF 35V

R1

1k

C1

100nF

ENC1

J1 encoder, AS5600 module

R2

1k

C2

100nF

ENC2

J2 encoder, AS5600 module

R3

1k

C3

100nF

ENC3

J3 encoder, AS5600 module

R4

1k

C4

100nF

ENC4

J4 encoder, AS5600 module

R5

1k

C5

100nF

ENC5

J5 encoder, AS5600 module

FAN1

40 mm 5 V fan over driver row

F1

2 A

U6

XL4015 buck, set 6.0 V

F2

3 A

C7

1000uF

C8

100nF

R13

1k

J6

Tool plug, GX12 6-pin

MG996R gripper servo

Gripper power: 24 V bus -> F1 -> U6 buck (set 6.0 V with a meter BEFORE connecting the servo) -> F2 -> 6 V (orange) -> C7, C8, servo.

BILL OF MATERIALS (electrical parts; full list, links and prices in hardware/bom)

Ref	Qty	Part	Note
U1	1	Raspberry Pi 5 8GB	owned
PS2	1	Official Raspberry Pi 27 W USB-C supply	owned
U2	1	Arduino Mega 2560 R3	owned
PS1	1	24 V bench supply	owned
DRV1–DRV5	5	BIGTREETECH TMC2209 V1.3 + heatsink	
M1, M3	2	STEPPERONLINE 17HS19-1684S-PG5	J1, J3
M2	1	STEPPERONLINE 17HS19-1684S-PG14	J2
M4	1	STEPPERONLINE 17HS08-1004S	J4
M5	1	STEPPERONLINE 11HS12-0674S	J5
W7	1	MG996R servo	J6 gripper tool
J6	1	GX12 6-pin tool plug	
U6	1	XL4015 5 A buck, set 6.0 V	
ENC1–ENC5	5	AS5600 analog encoder module + diametric magnet	
F0–F2	3	Inline ATO fuse holders, 10 A, 2 A and 3 A fuses	
SW1	1	E-stop, 22 mm latching mushroom, 1 NC	
K1	1	24 V coil relay, normally open, with socket	
D1	1	1N4007 flyback diode	
C7, C20–C24	6	Electrolytic 1000 uF 16 V (x1), 100 uF 35 V (x5)	
C1–C5, C8	6	Ceramic 100 nF	
R1–R5, R13, R21–R25, R31–R32	13	1k (x8), 10k (x5) metal film 1/4 W	
FAN1	1	40 mm 5 V fan over driver row	

NOTES
1. Mega pins: STEP D22–26, DIR D30–34, EN D36–40 (LOW = driver on; 10k pull-ups keep drivers OFF while the Mega boots), encoders A0–A4, gripper PWM D9, tool I2C D20–D21.
2. TMC2209 in UART mode: MS1/MS2 set each driver bus address (0–3); microsteps and current are set over UART by the firmware. CLK = GND selects the internal clock.
3. Pin names follow BIGTREETECH TMC2209 V1.2/V1.3. UART is on pin 4 (factory default). Check against the silkscreen of the modules you buy. The trimmer is not used in UART mode.
4. TMC2209 VM max is about 29 V; 24 V only. Never plug or unplug a motor with 24 V present. Verify coil pairs with an ohmmeter.
5. Modules sit in female headers on perfboard, see carrier/. Heatsink on every module. 40 mm fan over the row.
6. All grounds common. Never connect 24 V or 6 V to the Mega 5 V pins.

Module wiring — not a validated machine
RoboArm
Sheet: /
File: custom_arm.kicad_sch
Title: RoboArm — full wiring
Size: A1 | Date: 2026-11-02 | Rev: E
KiCad E.D.A. 9.0.9 | Id: 1/1